

Ex 1:Spring Data JPA - Quick Example

application.properties:

```
1# Spring Framework and application log
2logging.level.org.springframework=info
3logging.level.com.cognizant=debug
4
5# Hibernate logs
6logging.level.org.hibernate.SQL=trace
7logging.level.org.hibernate.type.descriptor.sql=trace
8
9# Log pattern
10logging.pattern.console=%d{dd-MM-yy} %d{HH:mm:ss.SSS} %-20.20thread %5p %-25.25logger{25} %25M %4L %m%n
11
12# Database configuration
13spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
14spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
15spring.datasource.username=root
16spring.datasource.password=PASSWORD_REDACTED
17
18# Hibernate configuration
19spring.jpa.hibernate.ddl-auto=validate
20spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQLDialect
```

Output:

```
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 9.881 s -- in com.cognizant.ormlearn.OrmLearnApplicationTests
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ orm-learn ---
[INFO] Building jar: C:\Users\anjan\workspace\orm-learn\target\orm-learn-0.0.1-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:4.1.0:repackage (repackage) @ orm-learn ---
[INFO] Replacing main artifact C:\Users\anjan\workspace\orm-learn\target\orm-learn-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested de
[INFO] The original artifact has been renamed to C:\Users\anjan\workspace\orm-learn\target\orm-learn-0.0.1-SNAPSHOT.jar.original
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 21.459 s
[INFO] Finished at: 2026-07-13T13:06:43+05:30
[INFO]
```

Country table creation:

```
1 • CREATE SCHEMA ormlearn;
2
3 • USE ormlearn;
4
5 • CREATE TABLE country (
6     co_code VARCHAR(2) PRIMARY KEY,
7     co_name VARCHAR(50)
8 );
9
10 • INSERT INTO country VALUES ('IN', 'India');
11 • INSERT INTO country VALUES ('US', 'United States of America');
```

Persistence Class - com.cognizant.ormlearn.model.Country

```
1 package com.cognizant.ormlearn.model;
2
3 import jakarta.persistence.Column;
4 import jakarta.persistence.Entity;
5 import jakarta.persistence.Id;
6 import jakarta.persistence.Table;
7
8 @Entity
9 @Table(name = "country")
10 public class Country {
11
12     @Id
13     @Column(name = "co_code")
14     private String code;
15
16     @Column(name = "co_name")
17     private String name;
18
19     public Country() {
20     }
21
22     public String getCode() {
23         return code;
24     }
25
26     public void setCode(String code) {
27         this.code = code;
28     }
29
30     public String getName() {
31         return name;
32     }
33
34     public void setName(String name) {
35         this.name = name;
36     }
37
38     @Override
39     public String toString() {
40         return "Country [code=" + code + ", name=" + name + "]";
41     }
42 }
```

Repository Class - com.cognizant.ormlearn.CountryRepository:

```
1 package com.cognizant.ormlearn.repository;
2
3 import org.springframework.data.jpa.repository.JpaRepository;
4 import org.springframework.stereotype.Repository;
5
6 import com.cognizant.ormlearn.model.Country;
7
8 @Repository
9 public interface CountryRepository extends JpaRepository<Country, String> {
10
11 }
```

Service Class - com.cognizant.ormlearn.service.CountryService:

```
1 package com.cognizant.ormlearn.service;
2
3 import java.util.List;
4
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.stereotype.Service;
7 import org.springframework.transaction.annotation.Transactional;
8
9 import com.cognizant.ormlearn.model.Country;
10 import com.cognizant.ormlearn.repository.CountryRepository;
11
12 @Service
13 public class CountryService {
14
15     @Autowired
16     private CountryRepository countryRepository;
17
18     @Transactional
19     public List<Country> getAllCountries() {
20         return countryRepository.findAll();
21     }
22 }
```

OrmLearnApplication.java:

```
1 package com.cognizant.ormlearn;
2
3 import java.util.List;
4
5 import org.springframework.boot.SpringApplication;
6 import org.springframework.boot.autoconfigure.SpringBootApplication;
7 import org.springframework.context.ApplicationContext;
8 import org.slf4j.Logger;
9 import org.slf4j.LoggerFactory;
10
11 import com.cognizant.ormlearn.model.Country;
12 import com.cognizant.ormlearn.service.CountryService;
13
14 @SpringBootApplication
15 public class OrmLearnApplication {
16
17     private static final Logger LOGGER = LoggerFactory.getLogger(OrmLearnApplication.class);
18     private static CountryService countryService;
19
20     public static void main(String[] args) {
21
22         ApplicationContext context = SpringApplication.run(OrmLearnApplication.class, args);
23
24         LOGGER.info("Inside main");
25
26         countryService = context.getBean(CountryService.class);
27
28         testGetAllCountries();
29     }
30
31     private static void testGetAllCountries() {
32
33         LOGGER.info("Start");
34
35         List<Country> countries = countryService.getAllCountries();
36
37         LOGGER.debug("countries={}", countries);
38
39         LOGGER.info("End");
40     }
41
42 }
```

```


:: Spring Boot ::                (v4.1.0)

13-07-26 12:45:41.420 restartedMain INFO c.c.o.OrmLearnApplication logStarting 54 Starting OrmLearnApplication using Java 17.0.19 with PID 12328 (C:
13-07-26 12:45:41.428 restartedMain DEBUG c.c.o.OrmLearnApplication logStarting 55 Running with Spring Boot v4.1.0, Spring v7.0.8
13-07-26 12:45:41.431 restartedMain INFO c.c.o.OrmLearnApplication logStartupProfileInfo 652 No active profile set, falling back to 1 default profile: "default
13-07-26 12:45:41.518 restartedMain INFO ertyDefaultsPostProcessor log 69 Devtools property defaults active! Set 'spring.devtools.add-proper
13-07-26 12:45:42.330 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 145 Bootstrapping Spring Data JPA repositories in DEFAULT mode.
13-07-26 12:45:42.332 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 213 Finished Spring Data repository scanning in 83 ms. Found 1 JPA rep
13-07-26 12:45:43.047 restartedMain INFO org.hibernate.orm.jpa logPersistenceUnitInformation 102 HHH0008540: Processing PersistenceUnitInfo [name: default]
13-07-26 12:45:43.197 restartedMain INFO org.hibernate.orm.core logVersion 41 HHH000001: Hibernate ORM core version 7.4.1.Final
13-07-26 12:45:43.991 restartedMain INFO SpringPersistenceUnitInfo addTransformer 129 No LoadTimeWeaver setup; ignoring JPA class transformer
13-07-26 12:45:44.051 restartedMain INFO c.z.h.HikariDataSource getConnection 109 HikariPool-1 - Starting...
13-07-26 12:45:44.700 restartedMain INFO c.z.h.pool.HikariPool checkFailFast 580 HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@e
13-07-26 12:45:44.707 restartedMain INFO c.z.h.HikariDataSource getConnection 122 HikariPool-1 - Start completed.
13-07-26 12:45:44.883 restartedMain WARN o.h.o.orm.deprecation constructDialect 152 HHH90000025: MySQLDialect does not need to be specified explicitly
13-07-26 12:45:44.957 restartedMain INFO o.h.o.connections.pooling logConnectionInfo 179 HHH10001005: Database info:

Database JDBC URL [jdbc:mysql://localhost:3306/ormlearn]
Database driver: MySQL Connector/J
Database dialect: MySQLDialect
Database version: 8.0.46
Default catalog/schema: ormlearn/undefined
Autocommit mode: undefined/unknown
Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown

13-07-26 12:45:47.914 restartedMain INFO org.hibernate.orm.core initiateService 53 HHH000489: No JTA platform available (set 'hibernate.transaction.j
13-07-26 12:45:48.141 restartedMain INFO rEntityManagerFactoryBean buildNativeEntityManagerFactory 464 Initialized JPA EntityManagerFactory for persistence unit 'd
13-07-26 12:45:48.472 restartedMain INFO .q.QueryEnhancerFactories <clinit> 48 Hibernate is in classpath; If applicable, HQL parser will be used.
13-07-26 12:45:49.490 restartedMain INFO c.c.o.OrmLearnApplication logStarted 60 Started OrmLearnApplication in 8.84 seconds (process running for 9
13-07-26 12:45:49.507 restartedMain INFO c.c.o.OrmLearnApplication main 24 Inside main
13-07-26 12:45:49.512 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 33 Start
13-07-26 12:45:50.339 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1_0.co_code,c1_0.co_name from country c1_0
13-07-26 12:45:50.522 restartedMain DEBUG c.c.o.OrmLearnApplication testGetAllCountries 37 countries=[Country {code=IN, name=India}, Country {code=US, name=U
13-07-26 12:45:50.524 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 39 End
13-07-26 12:45:50.551 licationsShutdownHook INFO rEntityManagerFactoryBean destroy 693 Closing JPA EntityManagerFactory for persistence unit 'default'
13-07-26 12:45:50.576 licationsShutdownHook INFO c.z.h.HikariDataSource close 349 HikariPool-1 - Shutdown initiated...
13-07-26 12:45:50.608 licationsShutdownHook INFO c.z.h.HikariDataSource close 351 HikariPool-1 - Shutdown completed.
```